

Chapter 3: Acids, Bases and Salts

3.1 Introduction

Substances around us can be classified as acidic, basic, or neutral based on their chemical properties. An acid is a substance that releases hydrogen ions (H^+) when dissolved in water. Common examples include hydrochloric acid (HCl), sulfuric acid (H_2SO_4), and acetic acid found in vinegar.

3.2 Properties of Acids

Acids taste sour, turn blue litmus paper red, and react with metals to produce hydrogen gas. They also react with bases in a process called neutralization, forming salt and water.

3.3 Properties of Bases

A base is a substance that releases hydroxide ions (OH^-) when dissolved in water. Bases taste bitter, feel slippery to touch, and turn red litmus paper blue. Common examples include sodium hydroxide ($NaOH$) and ammonia solution.

3.4 The pH Scale

The pH scale measures how acidic or basic a solution is, ranging from 0 to 14. A pH of 7 is neutral, values below 7 indicate acidity, and values above 7 indicate basicity. Pure water has a pH of exactly 7.

Figure 3.1: A diagram showing the pH scale from 0 to 14, with acids on the left, neutral at the center, and bases on the right.